

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
May-June 2016

Max. Marks: 100

Class: BE

Name of the Course: Soft Computing

Branch: COMP

Course Code: UCEE705

Duration: 3 Hrs.

Semester: MU-CBGS-VII

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1	Solve any four from the following	[20]
(a)	Define the terms core, width and support with suitable examples.	[05]
(b)	Explain with example Linearly separable and non-separable patterns.	[05]
(c)	Differentiate supervised and unsupervised learning?	[05]
(d)	What is Derivative free and derivative based optimization techniques?	[05]
(e)	Design OR gate using Mc-Culloch Pitts Neuron Model.	[05]
Q2	Solve any two from the following	[20]
(a)	For the given Fuzzy sets A & B, Find $A \cup B$, $A \cap B$, $A_{\alpha=0.5}$, $B_{\alpha=0.5}$ and $\bar{A}$ $A = \{0.2/a + 0.4/b + 1/c + 0.8/d + 0/e\}$ $B = \{0/a + 0.9/b + 0.3/c + 0.2/d + 0.1/e\}$	[10]
(b)	Design a Fuzzy system to predict price of the Resale cars. Assume Total Mileage and Date of Manufacturing as input variables(attributes) and Sale price as an output variable. Assume 3 descriptors(values of variables) for each input attributes and 5 descriptors for output variables. 1. Show clearly the units used for & universe of discourse of each variable. 2. Create Fuzzy Rulebase for Price prediction Problem. 3. Define fuzzy membership functions for all Input and Output variables	[02] [03] [05]
(c)	State with example extension Principle.	[10]
Q3	Solve any three of the following.	[18]
(a)	A neuron with 3 inputs has initial weight vector $W = [0.1 \ 0.3 \ -0.2]$. Calculate output for bipolar sigmoidal function for the input patterns $X1 = [0.8 \ 0.6 \ 0.4]$	[06]
(b)	Derive identity for bipolar continues function : $f^*(net) = O(1 - O^2)/2$	[06]
(c)	Determine the weights after two iterations for Hebbian learning of a single neuron network starting with initial weights $w = [1, -1]$, inputs as $X1 = [1, -2]$, $X2 = [2, 3]$, $X3 = [1, -1]$ and $c = 1$. Use Bipolar binary activation function	[06]
Q4	Solve any two from the following	[20]
(a)	Explain Travelling salesman problem using simulated Annealing.	[10]
(b)	What is RBFN? Compare RBF with MLP.	[10]
(c)	Explain Genetic algorithm. Show in detail all genetic operations. How it overcomes local minima?	[10]

Q5	Write short note on any one .	[10]
(a)	Printed Character Recognition.	[10]
(b)	Automobile Fuel Efficiency Prediction	[10]
Q6	Write short note on any two .	[12]
(a)	Learning Vector Quantization	[06]
(c)	Architecture of ANFIS	[06]